

Repeat the same argument at C , D , E , and on: each centripetal impulse Cc , always parallel to the radius SB , preserves the swept area, so $SCD = SBC$, $SDE = SCD$, and every triangle equals every other. By composition the whole swept region $SADFS$ grows exactly as the time elapses. Now let the number of triangles increase without bound and their breadth shrink to nothing: by Lemma V, Cor. 4 the ultimate perimeter ADF becomes a smooth curve, the pulsing thumps merge into a force acting *continually*, and the equal-area rule passes unbroken to the limit — $\frac{dA}{dt} = \text{const.}$ This holds for *any* central force, however it varies with distance: the force sculpts the *shape* of the orbit, but never touches the sacred equal-area law. *Q.E.D.*